SOFTWARE REQUIREMENTS SPECIFICATION

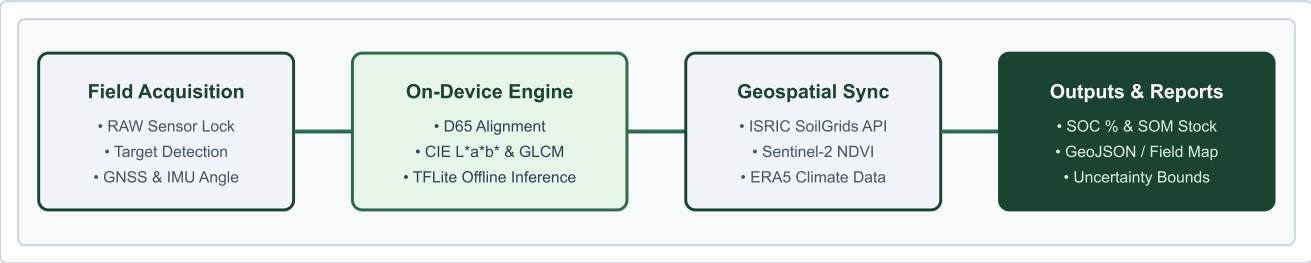
Technical Blueprint & Domain Reference: Smartphone Soil Carbon Estimation Engine

DOCUMENT ID	TARGET STANDARD	VERSION	CLASSIFICATION
SRS-SOC-2026-V1	ISO 10694 / USDA	2.4.0 (Final)	Technical Reference

1. Executive Overview & System Scope

This Software Requirements Specification (SRS) defines the operational, mathematical, and algorithmic requirements for the **Smartphone-Based Soil Organic Carbon (SOC) System**. The platform transforms commodity mobile devices into point-of-use optical soil analyzers. By capturing uncompressed camera sensor frames alongside a physical calibration target, the system isolates humic chromophore absorption in visible light (400–700 nm) and pairs it with spatial environmental covariates to deliver rapid, laboratory-grade SOC and SOM estimations.

2. System Operational Architecture



3. User Roles & System Stakeholders

USER CLASS	PRIMARY OPERATIONAL RESPONSIBILITIES	SYSTEM INTERFACES USED
Field Technician	Captures soil imagery, aligns optical calibration cards, tags spatial locations.	Mobile Capture UI, Camera Angle Alignment Guide.
Agronomist	Reviews carbon estimations, sets regional calibration overlays, exports reports.	Field Dashboard, GeoJSON/CSV Data Exporter.
Data Scientist	Trains regional machine learning models, manages cloud spectral databases.	Model Deployment Ops, SoilGrids API Sync.

4. Functional Requirements Specification

The software system must enforce hardware control, real-time image analytics, spatial metadata acquisition, and machine learning inference according to the functional modules specified below.

Module 1: Optical Capture & Hardware Control

- **FR-1.1 (Camera Parameter Lock):** The system shall programmatically lock ISO, shutter speed, manual focus distance, and turn off hardware auto-white-balance.
- **FR-1.2 (Bayer RAW Capture):** The mobile application shall store uncompressed RAW (Android DNG or iOS Bayer RAW) sensor frames directly alongside compressed JPEG assets.
- **FR-1.3 (Inclinometer Gate):** Image acquisition shall be inhibited if IMU pitch/roll sensors measure an off-axis tilt angle exceeding 10° from normal.

5. Hardware Sensors & Software API Matrix

COMPONENT	TYPE	TECHNICAL FUNCTION	API INTEGRATION
CMOS Camera	Hardware Sensor	Captures chromophore reflection & soil aggregate texture.	Camera2 / AVFoundation
GNSS / GPS	Hardware Sensor	Tag sample coordinates, elevation, and location accuracy.	LocationServices / CoreLocation
IMU Motion	Hardware Sensor	Validates perpendicular camera geometry (90° target alignment).	SensorManager / CoreMotion
BLE Spectrometer	External Device	Captures NIR molecular overtones (900–2500 nm).	Android Bluetooth LE API
Earth Engine API	Cloud Service	Fetches satellite vegetation index (NDVI) & surface climate.	REST / Google Earth Engine

6. Non-Functional System Requirements

- **NFR-1 (Inference Latency):** On-device Machine Learning inference (SOC % prediction from a single captured sample) shall execute in less than 2.0 seconds on mid-tier mobile hardware.
- **NFR-2 (Offline Cache Capacity):** The mobile app shall securely cache up to 1,000 un-synced field samples locally in an encrypted database (SQLite/Room) during extended offline field operations.
- **NFR-3 (Data Cryptography):** All local database entries, GPS tags, and raw image matrices must be encrypted at rest using AES-256 standard encryption.
- **NFR-4 (Provenance Integrity):** Every generated SOC prediction must attach a cryptographic hash of the raw image, camera exposure parameters, and model identifier for auditability.

7. Domain Science & Soil Physics Fundamentals

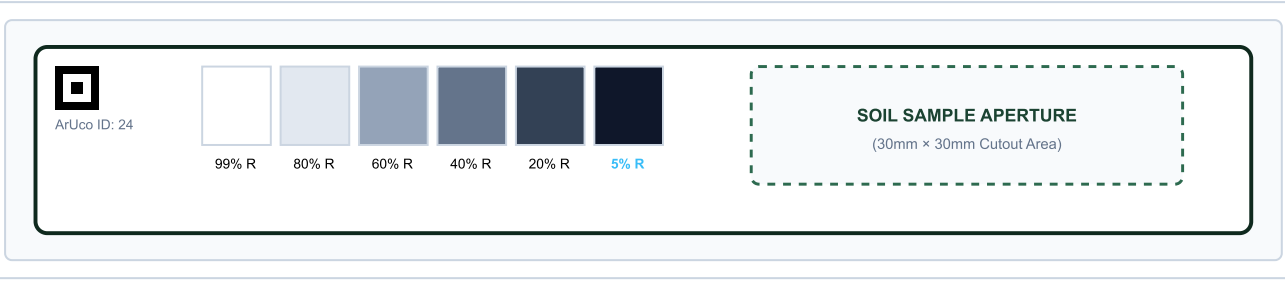
Soil Organic Carbon (SOC) is the primary component of Soil Organic Matter (SOM). Organic humic fractions (humic and fulvic acids) act as dark chromophores in visible spectrum wavelengths. As SOC concentration increases, total spectral reflectance decreases sharply in the 400–700 nm range.

Universal Carbon Conversion Standard (Van Bemmelen Factor):

$$SOM\ (\%) = SOC\ (\%) \times 1.724$$

Where SOC represents elemental organic carbon determined via dry combustion (ISO 10694).

8. Optical Calibration Target Specifications



- **Card Substrate:** Matte non-reflective synthetic polymer substrate (Yupo paper/polycarbonate); resistant to moisture, soil slurry, and UV degradation.
- **Fiducial Tracking:** Four corner ArUco/AprilTag matrices enable real-time sub-pixel perspective correction, distance estimation, and rotation alignment.
- **Illuminant Normalization:** The 6-patch diffuse grayscale array provides known reflectance baselines (99% down to 5%) to map raw image pixels to CIE D65 illuminant space.

9. Data Pipeline & Machine Learning Strategy

The analytical core utilizes a multi-stage feature extraction engine. Device-independent CIE L*a*b* coordinates isolate lightness (L*) as the primary SOC proxy. Texture features are extracted using Gray-Level Co-occurrence Matrices (GLCM), while External Parameter Orthogonalization (EPO) mitigates soil moisture interference.

Feature Vector Engineering Array:

```
Vector X = [ L*, a*, b*, Munsell_Value, GLCM_Contrast, GLCM_Energy, NDVI_Sat,
Clay_Content ]
```

10. Reference Databases & Benchmarks

DATASET / STANDARD	SOURCE ORGANIZATION	COVERAGE & ANALYTICAL UTILITY
LUCAS Soil Library	European Commission (ESDAC)	20,000+ topsoil samples across Europe with measured SOC and spectra (400–2500 nm).
KSSL Soil Database	USDA-NRCS (United States)	National pedon registry with laboratory dry combustion and MIR/VNIR diffuse reflectance.
ISO 10694	International Standards Org	Gold standard elemental dry combustion baseline for laboratory validation models.

11. Academic Foundation & Primary Citations

1. **Viscarra Rossel, R. A., et al. (2016).** *A global spectral library in the visible near-infrared and mid-infrared regions.* Earth-Science Reviews, 155, 35–67.
2. **Stiglitz, R., et al. (2017).** *Estimation of soil organic carbon and total nitrogen using color digital imaging.* Geoderma, 286, 12–23.
3. **Aitkenhead, M. J., et al. (2013).** *Smart phone app for estimation of soil properties from digital images.* Computers and Electronics in Agriculture, 97, 66–74.

Approved By: Lead Systems Architect **Date:** October 2026 **Status:** READY FOR IMPLEMENTATION